

T3 matched EFT Lagrangian terms: EFT_2_after_S1_S2

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out S1, S2. Remaining active T3 fields: none.

Field combinations 1 of 2

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$H \ell e$	$F H \ell$
cont.								$+\frac{2h\lambda_{T3}L_{a3,i1}}{\sqrt{3}M_F}CPR\overline{L_{a4,i2}}^TY_{1,i1,r1}Y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger$	$+\frac{3}{8}\sqrt{3h\overline{y_{2,i3,r1}L_{a1,i2}}}P_R e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	
								$+\frac{h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}{4\epsilon}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	$+\frac{h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}{4\epsilon}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	
cont.								$+\frac{3}{8}h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	$+\frac{3}{8}h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	
								$+\frac{\sqrt{3h\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}}{4\epsilon}P_L L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger$	$+\frac{\sqrt{3h\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}}{4\epsilon}P_L L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger$	
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{2h\overline{y_{1,i1,r1}Y_{2,i2,r1}L_{a4,i2}}}{\sqrt{3}\epsilon M_F}^TCPL_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}$	$+\frac{h\overline{y_{1,i3,r1}L_{a1,i2}}}{4\epsilon}P_R e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}$	$+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{1,i1,r1}Y_{2,i2,r1}L_{a4,i2}}}{\sqrt{3}M_F}^TCPL_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}$
								$+\frac{2h\overline{y_{1,i1,r1}Y_{2,i2,r1}L_{a4,i2}}}{\sqrt{3}M_F}^TCPL_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}$	$+\frac{3}{8}h\overline{y_{1,i3,r1}L_{a1,i2}}P_R e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}$	$+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\lambda_{T3}L_{a3,i1}CPR\overline{L_{a4,i2}}^TY_{1,i1,r1}Y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2h\lambda_{T3}L_{a3,i1}}{\sqrt{3}\epsilon M_F}CPR\overline{L_{a4,i2}}^TY_{1,i1,r1}Y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger$	$+\frac{\sqrt{3h\overline{y_{2,i3,r1}L_{a1,i2}}}}{4\epsilon}P_R e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	
								$+\frac{3}{8}\sqrt{3h\overline{y_{2,i3,r1}L_{a1,i2}}}P_R e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	$+\frac{3}{8}\sqrt{3h\overline{y_{2,i3,r1}L_{a1,i2}}}P_R e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	
cont.								$+\frac{h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}{4\epsilon}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	$+\frac{h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}{4\epsilon}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	
								$+\frac{3}{8}h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	$+\frac{3}{8}h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	
cont.								$+\frac{\sqrt{3h\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}}{4\epsilon}P_L L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger$	$+\frac{\sqrt{3h\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}}{4\epsilon}P_L L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger$	
								$+\frac{3}{8}\sqrt{3h\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}P_L L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger$	$+\frac{3}{8}\sqrt{3h\overline{y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}P_L L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger$	
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{2h\overline{y_{1,i1,r1}Y_{2,i2,r1}L_{a4,i2}}}{\sqrt{3}\epsilon M_F}^TCPL_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}$	$+\frac{h\overline{y_{1,i3,r1}L_{a1,i2}}}{4\epsilon}P_R e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}$	$+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\overline{y_{1,i1,r1}Y_{2,i2,r1}L_{a4,i2}}}{\sqrt{3}M_F}^TCPL_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}$
								$+\frac{2h\overline{y_{1,i1,r1}Y_{2,i2,r1}L_{a4,i2}}}{\sqrt{3}M_F}^TCPL_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}$	$+\frac{3}{8}h\overline{y_{1,i3,r1}L_{a1,i2}}P_R e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}$	$+\frac{2h\log\left(\frac{\hat{\mu}^2}{M_F^2}\right)\lambda_{T3}L_{a3,i1}CPR\overline{L_{a4,i2}}^TY_{1,i1,r1}Y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2h\lambda_{T3}L_{a3,i1}}{\sqrt{3}\epsilon M_F}CPR\overline{L_{a4,i2}}^TY_{1,i1,r1}Y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger$	$+\frac{h\overline{y_{2,i3,r1}L_{a1,i2}}}{4\epsilon}P_R e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	
								$+\frac{3}{8}h\overline{y_{2,i3,r1}L_{a1,i2}}P_R e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	$+\frac{3}{8}h\overline{y_{2,i3,r1}L_{a1,i2}}P_R e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	
cont.								$+\frac{h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}{4\epsilon}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	$+\frac{h\overline{y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}}}{4\epsilon}P_L L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	

[illegible]

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell \ell$	$H \ell e$	$F H \ell$	
<i>cont.</i>								$+\frac{3}{8}\sqrt{3h\overline{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
								$-\frac{2h\lambda_{T3}L_{a_3,i_1}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$	$+\frac{\sqrt{3h\overline{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
<i>cont.</i>								$+\frac{3}{8}\sqrt{3h\overline{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{\sqrt{3h\overline{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$			
									$+\frac{3}{8}\sqrt{3h\overline{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$		
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$	$+\frac{\sqrt{3h\overline{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	
								$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$+\frac{3}{8}\sqrt{3h\overline{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$	
<i>cont.</i>								$-\frac{2h\lambda_{T3}L_{a_3,i_1}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon M_F}$	$+\frac{\sqrt{3h\overline{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$		
									$+\frac{3}{8}\sqrt{3h\overline{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$		
<i>cont.</i>								$-\frac{2h\lambda_{T3}L_{a_3,i_1}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$	$+\frac{\sqrt{3h\overline{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
									$+\frac{3}{8}\sqrt{3h\overline{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$		
<i>cont.</i>									$+\frac{\sqrt{3h\overline{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
									$+\frac{3}{8}\sqrt{3h\overline{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$		
T3-C	0	2	0	2	1	3	$\frac{1}{2}$	$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$	$+\frac{\sqrt{3h\overline{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	
								$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$+\frac{3}{8}\sqrt{3h\overline{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$	
<i>cont.</i>								$-\frac{2h\lambda_{T3}L_{a_3,i_1}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon M_F}$	$+\frac{\sqrt{3h\overline{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$		
									$+\frac{3}{8}\sqrt{3h\overline{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$		
<i>cont.</i>								$-\frac{2h\lambda_{T3}L_{a_3,i_1}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$	$+\frac{\sqrt{3h\overline{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
									$+\frac{3}{8}\sqrt{3h\overline{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$		
<i>cont.</i>									$+\frac{\sqrt{3h\overline{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
									$+\frac{3}{8}\sqrt{3h\overline{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$		
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$	$+\frac{\sqrt{3h\overline{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	
								$-\frac{2h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}L_{a_4,i_2}^TCPL_{a_2,i_1}H_{a_1}H_{a_3}\lambda_{T3}\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$+\frac{3}{8}\sqrt{3h\overline{y}_{1,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{2h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{\lambda_{T3}L_{a_3,i_1}}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$	
<i>cont.</i>								$-\frac{2h\lambda_{T3}L_{a_3,i_1}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3\epsilon M_F}$	$+\frac{\sqrt{3h\overline{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$		
									$+\frac{3}{8}\sqrt{3h\overline{y}_{2,i_3,r_1}}\overline{L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$		
<i>cont.</i>								$-\frac{2h\lambda_{T3}L_{a_3,i_1}CP_R\overline{L_{a_4,i_2}}^TY_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}^\dagger H_{a_2}^\dagger}{3M_F}$	$+\frac{\sqrt{3h\overline{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
									$+\frac{3}{8}\sqrt{3h\overline{y}_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$		
<i>cont.</i>									$+\frac{\sqrt{3h\overline{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger}{4\epsilon}$		
									$+\frac{3}{8}\sqrt{3h\overline{y}_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$H \ell e$	$F H \ell$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$-\frac{2h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3\epsilon M_F}$	$\frac{\sqrt{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2h \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3M_F}$
								$-\frac{2h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3M_F}$	$+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2h \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3M_F}$
								$-\frac{2h\lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3\epsilon M_F}$	$+\frac{\sqrt{3h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	
cont.								$-\frac{2h\lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3M_F}$	$+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{\sqrt{3h\bar{y}_{2,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3\epsilon} M_F}$	$\frac{\sqrt{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2h \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3M_F}}$
								$+\frac{2h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3M_F}}$	$+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2h \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3M_F}}$
								$+\frac{2h\lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3\epsilon} M_F}$	$+\frac{h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_2}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
cont.								$+\frac{2h\lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3M_F}}$	$+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{h\bar{y}_{2,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}}{4\epsilon} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{3}{8}h\bar{y}_{2,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
T3-D	-1	3	$-\frac{1}{2}$	1	$\frac{1}{2}$	2	0	$+\frac{2h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3\epsilon} M_F}$	$\frac{\sqrt{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2h \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3M_F}}$
								$+\frac{2h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3M_F}}$	$+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2h \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3M_F}}$
								$+\frac{2h\lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3\epsilon} M_F}$	$+\frac{h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_2}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
cont.								$+\frac{2h\lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3M_F}}$	$+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{h\bar{y}_{2,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}}{4\epsilon} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{3}{8}h\bar{y}_{2,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3\epsilon} M_F}$	$\frac{\sqrt{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$+\frac{2h \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \bar{y}_{1,i_1,r_1} \bar{y}_{2,i_2,r_1} \bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3M_F}}$
								$+\frac{2h\bar{y}_{1,i_1,r_1}\bar{y}_{2,i_2,r_1}\bar{L}_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3M_F}}$	$+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_3,r_1}\bar{L}_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2h \log\left(\frac{\hat{\mu}^2}{M_F^2}\right) \lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3M_F}}$
								$+\frac{2h\lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3\epsilon} M_F}$	$+\frac{h\bar{y}_{2,i_3,r_1}\bar{L}_{a_1,i_2}}{4\epsilon} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	
cont.								$+\frac{2h\lambda_{T3} \bar{L}_{a_3,i_1} C P_R \bar{L}_{a_4,i_2}{}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3M_F}}$	$+\frac{\sqrt{3h\bar{y}_{1,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3h\bar{y}_{1,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{h\bar{y}_{2,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2}}{4\epsilon} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	
									$+\frac{3}{8}h\bar{y}_{2,i_1,r_1}\bar{Y}_{e,i_3,i_2}\bar{e}_{i_2} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}{}^\dagger$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell$	$H \ell e$	$F H \ell$
cont.									$+\frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$	
T3-D	1	3	$\frac{1}{2}$	1	$\frac{3}{2}$	2	1	$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8} \sqrt{3}\hbar \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8} \hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8} \sqrt{3}\hbar \overline{y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$ $+\frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$	
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8} \sqrt{3}\hbar \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8} \hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$+\frac{2\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$
cont.								$+\frac{2\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\sqrt{3}M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8} \sqrt{3}\hbar \overline{y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$ $+\frac{\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8} \hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$	
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\epsilon M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8} \sqrt{3}\hbar \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8} \sqrt{3}\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$
cont.								$-\frac{2\sqrt{\frac{2}{3}}\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$ $+\frac{3}{8} \sqrt{3}\hbar \overline{y_{1,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$ $+\frac{\sqrt{3}\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8} \sqrt{3}\hbar y_{2,i_1,r_1} \overline{Y_{e,i_3,i_2} e_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$	
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\epsilon M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \lambda_{T3} \overline{L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{\epsilon M_F}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8} \sqrt{3}\hbar \overline{y_{1,i_3,r_1} \overline{L_{a_1,i_2}}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$ $+\frac{\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{\sqrt{3}\hbar y_{2,i_3,r_1} \overline{L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_{T3} \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{\lambda_{T3} L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}^\dagger H_{a_2}^\dagger}{M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H \ell \ell$	$H \ell e$	$F H \ell$
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{M_F}$	$+\frac{3}{8}\sqrt{3h\overline{y_{2,i3,r1}}\overline{L_{a1,i2}}}PR e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	
									$+\frac{\sqrt{3h}y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}PL L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger}{4\epsilon}$	
cont.									$+\frac{3}{8}\sqrt{3h\overline{y_{1,i1,r1}}\overline{Y_{e,i3,i2}}\overline{e_{i2}}}PL L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	
									$+\frac{\sqrt{3h}y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}PL L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger}{4\epsilon}$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i1,r1}}\overline{y_{2,i2,r1}}\overline{L_{a4,i2}}^T CPL L_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}}{\epsilon M_F}$	$+\frac{\sqrt{3h}y_{1,i3,r1}\overline{L_{a1,i2}}PR e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}}{4\epsilon}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i1,r1}}\overline{y_{2,i2,r1}}\overline{L_{a4,i2}}^T CPL L_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}}{M_F}$
								$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i1,r1}}\overline{y_{2,i2,r1}}\overline{L_{a4,i2}}^T CPL L_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}}{M_F}$	$+\frac{3}{8}\sqrt{3h\overline{y_{1,i3,r1}}\overline{L_{a1,i2}}}PR e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{M_F}$
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{\epsilon M_F}$	$+\frac{\sqrt{3h}y_{2,i3,r1}\overline{L_{a1,i2}}PR e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3h\overline{y_{2,i3,r1}}\overline{L_{a1,i2}}}PR e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{M_F}$	$+\frac{\sqrt{3h}y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}PL L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3h\overline{y_{1,i1,r1}}\overline{Y_{e,i3,i2}}\overline{e_{i2}}}PL L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	
cont.									$+\frac{\sqrt{3h}y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}PL L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3h\overline{y_{2,i1,r1}}\overline{Y_{e,i3,i2}}\overline{e_{i2}}}PL L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger$	
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i1,r1}}\overline{y_{2,i2,r1}}\overline{L_{a4,i2}}^T CPL L_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}}{\epsilon M_F}$	$+\frac{\sqrt{3h}y_{1,i3,r1}\overline{L_{a1,i2}}PR e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}}{4\epsilon}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i1,r1}}\overline{y_{2,i2,r1}}\overline{L_{a4,i2}}^T CPL L_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}}{M_F}$
								$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i1,r1}}\overline{y_{2,i2,r1}}\overline{L_{a4,i2}}^T CPL L_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}}{M_F}$	$+\frac{3}{8}\sqrt{3h\overline{y_{1,i3,r1}}\overline{L_{a1,i2}}}PR e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{M_F}$
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{\epsilon M_F}$	$+\frac{\sqrt{3h}y_{2,i3,r1}\overline{L_{a1,i2}}PR e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3h\overline{y_{2,i3,r1}}\overline{L_{a1,i2}}}PR e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{M_F}$	$+\frac{\sqrt{3h}y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}PL L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3h\overline{y_{1,i1,r1}}\overline{Y_{e,i3,i2}}\overline{e_{i2}}}PL L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	
cont.									$+\frac{\sqrt{3h}y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}PL L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3h\overline{y_{2,i1,r1}}\overline{Y_{e,i3,i2}}\overline{e_{i2}}}PL L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i1,r1}}\overline{y_{2,i2,r1}}\overline{L_{a4,i2}}^T CPL L_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}}{\epsilon M_F}$	$+\frac{\sqrt{3h}y_{1,i3,r1}\overline{L_{a1,i2}}PR e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}}{4\epsilon}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i1,r1}}\overline{y_{2,i2,r1}}\overline{L_{a4,i2}}^T CPL L_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}}{M_F}$
								$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i1,r1}}\overline{y_{2,i2,r1}}\overline{L_{a4,i2}}^T CPL L_{a2,i1}H_{a1}H_{a3}\lambda_{T3}\epsilon_{a1,a2}\epsilon_{a3,a4}}{M_F}$	$+\frac{3}{8}\sqrt{3h\overline{y_{1,i3,r1}}\overline{L_{a1,i2}}}PR e_{i1}H_{a1}y_{1,i2,r1}Y_{e,i3,i1}$	$-\frac{2\sqrt{\frac{2}{3}}h\log\left(\frac{\mu^2}{M_F^2}\right)\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{M_F}$
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{\epsilon M_F}$	$+\frac{\sqrt{3h}y_{2,i3,r1}\overline{L_{a1,i2}}PR e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3h\overline{y_{2,i3,r1}}\overline{L_{a1,i2}}}PR e_{i1}H_{a1}y_{2,i2,r1}Y_{e,i3,i1}$	
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\lambda_{T3}L_{a3,i1}CP_R\overline{L_{a4,i2}}^T y_{1,i1,r1}y_{2,i2,r1}\epsilon_{a1,a3}\epsilon_{a2,a4}H_{a1}^\dagger H_{a2}^\dagger}{M_F}$	$+\frac{\sqrt{3h}y_{1,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}PL L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3h\overline{y_{1,i1,r1}}\overline{Y_{e,i3,i2}}\overline{e_{i2}}}PL L_{a1,i1}y_{1,i3,r1}H_{a1}^\dagger$	
cont.									$+\frac{\sqrt{3h}y_{2,i1,r1}Y_{e,i3,i2}\overline{e_{i2}}PL L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger}{4\epsilon}$	
									$+\frac{3}{8}\sqrt{3h\overline{y_{2,i1,r1}}\overline{Y_{e,i3,i2}}\overline{e_{i2}}}PL L_{a1,i1}y_{2,i3,r1}H_{a1}^\dagger$	

Field combinations 2 of 2

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	e
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
cont.								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{2,i_2,r_1} Y_{e,i_3,i_1}$			
T3-A	-1	1	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-A	1	1	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-A	2	1	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-B	-2	2	-1	2	0	1	$-\frac{1}{2}$	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{1,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{2,i_1,r_1} Y_{e,i_3,i_2} \bar{e}_{i_2}} P_L L_{a_1,i_1} y_{2,i_3,r_1} H_{a_1}^\dagger$			
								$+\frac{1}{4}h \log \left(\frac{\bar{\mu}^2}{M_F^2} \right) \overline{y_{1,i_3,r_1} L_{a_1,i_2}} P_R e_{i_1} H_{a_1} y_{1,i_2,r_1} Y_{e,i_3,i_1}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell e$
T3-B	0	2	0	2	1	1	$\frac{1}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-B	2	2	1	2	2	1	$\frac{3}{2}$	$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-2	2	-1	2	0	3	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}\bar{e}_{i_2}}P_L L_{a_1,i_1}y_{2,i$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F H \ell e$
T3-C	2	2	1	2	2	3	$\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}PRe_{i_1}H_{a_1}y_{1,i_2,r_1}Ye_{i_3,i_1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}PRe_{i_1}H_{a_1}y_{2,i_2,r_1}Ye_{i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Ye_{i_3,i_2}\bar{e}_{i_2}}PL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Ye_{i_3,i_2}\bar{e}_{i_2}}PL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}PRe_{i_1}H_{a_1}y_{1,i_2,r_1}Ye_{i_3,i_1}$ $+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}PRe_{i_1}H_{a_1}y_{2,i_2,r_1}Ye_{i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Ye_{i_3,i_2}\bar{e}_{i_2}}PL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{1}{4}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Ye_{i_3,i_2}\bar{e}_{i_2}}PL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_2}}PRe_{i_1}H_{a_1}y_{1,i_2,r_1}Ye_{i_3,i_1}$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_2}}PRe_{i_1}H_{a_1}y_{2,i_2,r_1}Ye_{i_3,i_1}$
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Ye_{i_3,i_2}\bar{e}_{i_2}}PL_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$ $+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Ye_{i_3,i_2}\bar{e}_{i_2}}PL_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	e
T3-E	-1	3	$-\frac{1}{2}$	3	$\frac{1}{2}$	2	0	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	1	3	$\frac{1}{2}$	3	$\frac{3}{2}$	2	1	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{1,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}{\overline{y_{2,i_3,r_1}L_{a_1,i_2}}}P_R e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$			
cont.								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{1,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}^\dagger$			
								$+\frac{1}{4}\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\frac{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}{\overline{y_{2,i_1,r_1}Y_{e,i_3,i_2}}}\overline{e_{i_2}}P_L L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}^\dagger$			